the universe started out right after the Big Bang in an orderly state, and entropy has naturally been increasing ever since, and so we have time's arrow. We don't really know why the early universe had such a low entropy, though some of us have ideas."

"And this is relevant because . . ."

"Because for Everettians, the explanation of the quantum arrow of time is the same as that of the entropic arrow of time: the initial conditions of the universe. Branching happens when systems become entangled with the environment and decohere, which unfolds as time moves toward the future, not the past. The number of branches of the wave function, just like the entropy, only increases with time. That means that the number of branches was relatively small to begin with. In other words, that there was a relatively low amount of entanglement between various systems and the environment in the far past. As with entropy, this is an initial condition we impose on the state of the universe, and at the present time we don't know for sure why it was the case."

"Okay," said her father. "It's good to admit what we don't know. We explain the arrow of time, at least according to the current state of the art, by appeal to special initial conditions in the past. Is it a single condition that explains both the thermodynamic arrow and the quantum arrow, or is that just an analogy?"

"I think it's more than an analogy, but to be honest, this is a subject that could probably use a bit more rigorous investigation," Alice replied. "There certainly seems to be a connection. Entropy is related to our ignorance. If a system has low entropy, there are relatively few microscopic configurations that would look that way, so we know a lot about it just from its macroscopically observable features; if it has high entropy, we know relatively little. John von Neumann realized that we can say something similar about entangled quantum systems. If a system is completely unentangled with anything else, we can safely talk about its wave function in isolation from the rest of the world. But when it is entangled, the individual wave function is undefined, and we can only talk about the wave function for the combined system."

Her father brightened. "Von Neumann was a brilliant guy, a real hero. There were an amazing number of Hungarian physicists who emigrated to the US—Szilard, Wigner, Teller—but he was the top. I do vaguely remember that he derived a formula for entropy."